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Integrated profiling reveals polarity protein dysregulation during oral cancer progression

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Malignant transformation of oral precancerous lesions is a multistep process intricately linked to the disruption of epithelial cell polarity and activation of the epithelial–mesenchymal transition (EMT) program. This study provides an integrated analysis of the polarity regulators PAR3, SCRIBBLE, and DLG7, elucidating their differential expression across normal oral mucosa (NOM), oral epithelial dysplasia (OED), and oral squamous cell carcinoma (OSCC). By combining histopathological evaluation, immunohistochemical profiling, and whole-transcriptome sequencing, this work offers novel insights into polarity disruption as a driving mechanism in oral tumorigenesis. Formalin-fixed, paraffin-embedded (FFPE) tissue specimens were obtained from 50 habitual tobacco users from West Bengal, India. Sections were stained with hematoxylin and eosin (H&E) for histopathological assessment and classification into normal oral mucosa (NOM), oral epithelial dysplasia (OED), and oral squamous cell carcinoma (OSCC) (well- and moderately-differentiated grades). Immunohistochemical (IHC) analysis was conducted to evaluate the expression and localization patterns of the polarity-associated proteins PAR3, SCRIBBLE, and DLG7. Complementing this, whole-transcriptome RNA sequencing was performed on biopsy specimens from an independent cohort of 25 oral cancer patients exhibiting both OED and OSCC lesions, enabling comparative gene expression profiling of the same polarity regulators. Statistical analysis using IBM SPSS (version 20.0) and Chi-square testing revealed a significant reduction or complete loss of PAR3, SCRIBBLE, and DLG7 expression in both oral epithelial dysplasia (OED) and oral squamous cell carcinoma (OSCC), compared to their moderate to strong expression in normal oral mucosa (NOM). This study reveals a striking decline in epithelial polarity protein expression from normal oral mucosa (NOM) to oral epithelial dysplasia (OED), followed by a modest resurgence in oral squamous cell carcinoma (OSCC). The strong concordance between immunohistochemical and transcriptomic profiles—with the exception of DLG7—highlights the disruption of cell polarity as an early and central molecular event in oral carcinogenesis. Collectively, the polarity regulators PAR3, SCRIBBLE, and DLG7 emerge as promising biomarkers for early malignant transformation in oral potentially malignant disorders (OPMDs) and as potential modulators of tumor initiation, progression, and invasive behavior.

Keywords Cell polarity, PAR3 (PARD3), SCRIBBLE (SCRIB), DLG7 (DLGAP5), Oral epithelial dysplasia (OED), Oral squamous cell carcinoma (OSCC), Epithelial–mesenchymal transition (EMT), Polarity disruption, Oral potentially malignant disorders (OPMDs), Immunohistochemistry, Transcriptomic profiling, Biomarker discovery, Polarity–EMT axis, Tumor initiation, Oral cancer progression, Precision oral oncology

Cell polarity is an evolutionarily conserved organizational principle that governs the asymmetric distribution of cellular constituents and is fundamental to epithelial integrity and function[1]. Distinct polarity complexes—including PAR and CRUMBS at the apical domain and SCRIBBLE and DLG at the basolateral domain—coordinate spatial compartmentalization through interactions with tight and adherens junctions[2–5]. Depending on cell type and physiological context, polarity assumes different configurations: epithelial cells exhibit apico–basal polarity, whereas motile cell types such as fibroblasts, immune cells, and cancer cells display front–rear polarity[2,3].

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Perturbation of polarity proteins disrupts epithelial cohesion and is closely linked to epithelial–mesenchymal transition (EMT), a pivotal step in tumor progression[6]. During EMT, epithelial cells lose apico–basal polarity and acquire mesenchymal, migratory characteristics; in some contexts, a "partial EMT" phenotype emerges, wherein both epithelial and mesenchymal traits coexist[7].

Despite its central role in other epithelial cancers, the contribution of polarity proteins to oral carcinogenesis remains largely unexplored. Oral squamous cell carcinoma (OSCC), which often arises from the potentially malignant disorder oral leukoplakia, provides a clinically relevant model to investigate polarity dysregulation[8–10]. Oral epithelial dysplasia (OED), the histopathological correlate of leukoplakia, remains the only established benchmark for predicting malignant transformation in oral potentially malignant disorders (OPMDs). Hallmarks of oral epithelial dysplasia (OED) include loss of basal cell polarity[11], histologically interpreted by nuclear misorientation reminiscent of reverse polarity—changes that parallel those observed in other epithelia, such as during ameloblast differentiation[12], columnar cell breast carcinoma[13], and papillary renal carcinoma[14]. In addition, cellular discohesion, another defining feature of OED, marked by the disruption of basal cell palisading and the appearance of discohesive cells, further signifies polarity dysregulation. Notably, both basal polarity loss and cellular discohesion directly implicate polarity mechanisms; however, such histomorphological surrogates may fail to capture the underlying molecular derangements, leaving the mechanistic and biological basis of OPMD progression incompletely understood. This knowledge may pave the way for identifying reliable biomarkers predictive of malignant transformation in OPMD, while also aiding in prognostication of OSCC. Moreover, the pivotal roles of polarity proteins in EMT could inform the development of novel therapeutic strategies[15].

To bridge this critical knowledge gap and to overcome the inherent subjectivity in interpreting dysplastic features, the present study investigates the molecular mechanisms of polarity disruption driving the transition from epithelial dysplasia to carcinoma. Guided by a biologically grounded rationale, we examined three cardinal polarity regulators: PAR3, a determinant of the apical domain, and SCRIBBLE and DLG7, integral to the basolateral polarity complex. Collectively, these proteins orchestrate epithelial architecture, signalling fidelity and growth control. Perturbations within these polarity modules not only dismantle apico–basal organization but also activate oncogenic signalling and epithelial–mesenchymal transition (EMT) programs, thereby linking polarity loss to malignant transformation. Their dysregulation has been recurrently implicated in the tumorigenesis of multiple epithelial malignancies, including cancers of the prostate, lung, and breast[16–23]. We evaluated their expression across normal oral mucosa (NOM), OED, and OSCC using histopathology and immunohistochemistry, and complemented these analyses with RNA-sequencing data from an independent cohort of 25 patients with leukoplakia (OED), OSCC, and matched normal tissues[24].

Materials and methods

Tissue samples

Biopsy tissues were obtained from patients attending the Department of Oral Pathology, Dr. R. Ahmed Dental College & Hospital, Kolkata, with immunohistochemical (IHC) analysis performed at the Chittaranjan National Cancer Institute. The study included normal oral mucosa (NOM), oral epithelial dysplasia (OED), and oral squamous cell carcinoma (OSCC). Fifty samples were selected by purposive and incidental sampling; patients with systemic illness, infections, or poorly differentiated OSCC were excluded. Normal controls were derived from benign lesions (fibroma, pyogenic granuloma), contralateral healthy mucosa in cancer patients.

Operational definitions

NOM

Oral mucosa with intact stratified squamous epithelium and normal lamina propria, free of dysplasia, inflammation, ulceration, or lesions, obtained from clinically healthy, non-lesional sites; used as control tissue in our study.

- OSCC: Ulcerated lesions with rolled margins, histologically confirmed by stromal invasion.
- Leukoplakia: White plaques resistant to scraping, often tobacco-associated, showing variable grades of epithelial dysplasia without invasion.

Immunohistochemistry (IHC)

Formalin-fixed, paraffin-embedded tissue blocks were sectioned at 4–5 μm and mounted on poly-L-lysine–coated slides. IHC was performed using rabbit polyclonal antibodies:

- PAR3: 1:400 dilution (Novus Biologicals, Centenial, Colorado, USA Cat. NBP-88861).
- SCRIBBLE: 1:200 dilution (Novus Biologicals, Centenial, Colorado, USA Cat. NBP2-15110).
- DLG7: 1:200 dilution (Novus Biologicals, Centenial, Colorado, USA Cat. NBP1-87976).

Staining was detected with Polymer HRP/DAB (ScyTek Laboratories, USA).

ScyTek DAB chromogen—technical details

- Active Ingredient: 3,3'-Diaminobenzidine (DAB).
- Formulation: Liquid concentrate.
- Working Dilution: 50 μL DAB Chromogen per 1 mL DAB Substrate.
- Reaction: Produces a brown, alcohol-insoluble precipitate in the presence of peroxidase.
- Shelf Life: 18 months from manufacture.

- Storage: 2–8 °C.
- Post-Mixing Stability: Up to 6 h.
- Control Tissue: Well-fixed frozen or FFPE tissue.

Slides were evaluated independently by three experienced oral pathologists. Staining intensity was graded: 0 = negative, 1 = weak, 2 = moderate, 3 = strong. An intensity scoring unit (ISU) was calculated as the ratio of observed weighted intensity to the maximum possible score.

RNA sequencing
Transcriptomic data from 25 independent patients (ENA Project ID: PRJEB42969) were analyzed. Pipeline included:

- Quality check: FastQC (https://github.com/s-andrews/FastQC), MultiQC[25];
- Alignment: STAR[26];
- QC: Qualimap[27];
- Quantification: HTSeq-count[28];
- Differential expression: DESeq2[29].

Significant changes were defined as log2 fold change > 1 or < − 1 with adjusted $p < 0.1$.

Statistical analysis
Descriptive data were presented as figures and bar charts. Protein expression differences among NOM, OED, and OSCC were tested with Chi-square. $p < 0.05$ was considered statistically significant. Analyses were conducted using IBM SPSS v20.0.

Results
Clinicopathological profile
Of 50 patients recruited, 10 were excluded due to inadequate follow-up or tissue quality. Thirty OSCC cases were retained for analysis. All participants were residents of West Bengal with a history of tobacco use (chewing, smoking, or both) (Table 1). A separate RNA-seq cohort comprised 25 patients with leukoplakia and concurrent OSCC (Table 2).

Histopathology
H&E-stained sections confirmed normal epithelium in NOM, variable grades of dysplasia in leukoplakia, and invasive carcinoma in OSCC (Fig. 1).

Immunohistochemistry
IHC demonstrated marked loss of polarity proteins from NOM to OED and NOM to OSCC.

- PAR3: In NOM, weak suprabasal expression was detected (ISU = 0.27). In OED, PAR3 expression was nearly absent (ISU = 0.067). Among OSCC samples, 23/30 were negative and 7 showed only weak expression (ISU = 0.078).
- SCRIBBLE: NOM showed strong basal and moderate suprabasal staining in 80% of cases (ISU = 0.94). Expression dropped markedly in OED (ISU = 0.1), with 70% negative. OSCC showed weak-to-absent staining in most cases (ISU = 0.14).
- DLG7: NOM displayed moderate-to-weak expression (ISU = 0.6). OED showed loss in 70% of cases (ISU = 0.1). In OSCC, 15/30 were negative, 14 weak, and 1 moderate (ISU = 0.18).

Representative staining patterns are shown in Figs. 2, 3 and 4.

Comparative immunohistochemical comparison identified significant loss of the expression of all three polarity-related proteins—PAR3, Scribble, and DLG7—from normal oral mucosa (NOM) to OED and OSCC tissues. Among these, Scribble had the greatest intensity in normal tissues (ISU 0.94), which fell steeply in OED (ISU 0.10) and in OSCC (ISU 0.14) tissues. Correspondingly, PAR3 expression was lowered from ISU 0.27 in NOM to ISU 0.07 in OED and ISU 0.08 in OSCC, and DLG7 expression was lowered from ISU 0.60 in NOM to ISU 0.10 and 0.14 in OED and OSCC, respectively. This progressive decrease in polarity protein expression with increasing histological severity highlights polarity deregulation as an earlier molecular event of oral carcinogenesis. This is represented in Fig. 5.

Transcriptomic analysis

- PARD3 and SCRIB were significantly downregulated in tumors (log2FC − 0.78 and − 0.69).
- DLGAP5 (encoding DLG7) was upregulated in tumors (log2FC + 1.19).
- In leukoplakia vs. NOM, PARD3 was significantly downregulated (log2FC − 0.61, $p = 0.000185$), while SCRIB trended downward and DLGAP5 upward without statistical significance.

Overall, the trajectory from normal → leukoplakia → carcinoma revealed consistent downregulation of PARD3 and SCRIB and progressive upregulation of DLGAP5 (Fig. 6).

S.no	Patient Id	Age and sex		Tobacco smoking with/without alcohol	Tobacco chewing with/without alcohol	Both tobacco smoking and chewing with or without alcohol	Tumour staging (clinical)	Histological subtype
1	RRB1	29	M	–	–	Yes	T2N1Mx	WDSCC
2	RRB2	38	M	–	–	Yes	T3N2cMx	WDSCC
3	RRB3	36	M	–	Yes	–	T2N1Mx	MDSCC
4	RRB4	32	M	–	Yes	–	T4aN2cMx	WDSCC
5	RRB5	32	M	–	Yes	–	T3N1Mx	WDSCC
6	RRB6	40	F	–	–	–	T3N2CMx	WDSCC
7	RRB7	38	M	–	–	Yes	T3N2AMx	MDSCC
8	RRB8	30	M	–	Yes	–	T2N1Mx	WDSCC
9	RRB9	38	M	Yes	–	–	T3N2CMx	MDSCC
10	RRB10	30	M	–	Yes	–	T3N2BMx	WDSCC
11	RRB11	30	M	–	Yes	–	T4AN2CMx	WDSCC
12	RRB12	30	M	–	–	Yes	T4AN3Mx	WDSCC
13	RRB13	40	M	–	–	Yes	T2N2AMx	WDSCC
14	RRB14	40	M	–	Yes	–	T1N0Mx	WDSCC
15	RRB15	29	M	Yes	–	–	T3N3Mx	WDSCC
16	RRA1	64	F	–	Yes	–	T4aN1Mx	WDSCC
17	RRA2	78	F	–	Yes	–	T3N2aMx	WDSCC
18	RRA3	70	M	–	–	–	T4aN2cMx	WDSCC
19	RRA4	47	F	–	Yes	–	T3N2bMx	MDSCC
20	RRA5	42	M	–	–	–	T4aN2bMx	WDSCC
21	RRA6	60	F	–	Yes	–	T4aN2cMx	MDSCC
22	RRA7	68	M	–	–	Yes	T3N1Mx	WDSCC
23	RRA8	47	F	–	Yes	–	T2N1Mx	WDSCC
24	RRA9	75	F	–	–	Yes	T3N3Mx	MDSCC
25	RRA10	48	M	Yes	–	–	T4aN2cMx	MDSCC
26	RRA11	55	M	–	–	Yes	T4aN2cMx	WDSCC
27	RRA12	60	M	–	Yes	–	T4aN3Mx	WDSCC
28	RRA13	44	M	Yes	–	–	T4aN3Mx	MDSCC
29	RRA14	60	M	–	Yes	–	T3N1Mx	WDSCC
30	RRA15	70	F	–	Yes	–	T4bN2cMx	WDSCC

Table 1. Clinicopathological characteristics of OSCC patients ($n = 30$) who were recruited for immunohistochemistry-based characterization of polarity-associated proteins.

Discussion

To elucidate how precancerous lesions transition into oral squamous cell carcinoma (OSCC), we interrogated polarity disruption by profiling three polarity-regulating proteins—PAR3, SCRIBBLE, and DLG7 (encoded by DLG5)—across Normal Oral Mucosa (NOM), Oral Epithelial Dysplasia (OED), and OSCC using immunohistochemistry (IHC) and transcriptomics.

In NOM, PAR3 expression was faint and restricted to suprabasal layers, consistent with its function in apical junctional complexes[15,30], whereas SCRIBBLE and DLG7 were widely expressed in basal and suprabasal layers, reflecting their localization at basolateral domains enriched in desmosomes[31,32]. The absence of PAR3 in basal progenitor cells is striking, suggesting compartment-specific polarity regulation that may shape epithelial homeostasis and regeneration[33].

In the oral epithelium, basal cell polarity cannot be reliably assessed by nuclear orientation, as basal cells are cuboidal with centrally placed nuclei[34]. Consequently, although loss of basal cell polarity and cellular discohesion are recognized histological features of dysplasia, their molecular underpinnings remain elusive[11]. The lack of mechanistic and objective biomarkers has confined dysplasia to a purely nebulous morphological predictor of malignant transformation in oral potentially malignant disorders (OPMDs). By interrogating polarity-regulating proteins, our study introduces a novel molecular framework that links epithelial architecture to dysplastic progression, offering a potential paradigm shift in biomarker discovery for oral cancer risk stratification[35].

We observed a marked reduction in PAR3, SCRIBBLE, and DLG7 expression in OED compared with NOM, with particularly pronounced depletion of SCRIBBLE and DLG7 within the basal cell layers—mirroring observations in gastric dysplasia, where polarity regulators such as LGL2 are similarly downregulated. These findings position polarity disruption as an early, causative event in malignant transformation rather than a secondary consequence of tumor progression[20]. Transcriptomic analysis corroborated the downregulation of PARD3 and SCRIB in dysplasia. In contrast, DLG5 mRNA was paradoxically expressed at almost the same level in normal (median TPM expression: 0.138) and OED (median TPM expression: 0.139) and markedly upregulated

Patient ID	Age	Sex	Tobacco chewing in form of "Khaini"	Tobacco chewing with betel leaf	Tobacco chewing (others)	Smoking (Bidi)	Alcohol	Histopathological grade (tumour)	Histopathological grade (leukoplakia)
2	53	M	No	Yes	No	Yes	No	MD	MID
3	40	F	No	Yes	No	No	No	MD	MID
4	32	M	Yes	No	No	Yes	Yes		
5	50	F	No	No	Yes	No	No	WD	MID
6	60	M	No	No	No	Yes	No	WD	MID
7	65	M	No	No	No	Yes	Yes	WD	MOD
8	51	M	No	No	No	Yes	No	WD	MOD
10	65	M	Yes	No	No	No	Yes	WD	MOD
13	50	F	Yes	No	Yes	No	No	WD	SD
17	53	M	Yes	No	Yes	Yes	No	WD	MID
18	58	M	No	No	Yes	Yes	No	WD	MID
19	40	O	Yes	No	Yes	No	No	WD	SD
23	45	M	No	Yes	No	No	No	WD	MID
27	50	M	No	Yes	No	Yes	No	EI	MOD
28	55	M	Yes	Yes	No	Yes	No	WD	SD
29	36	M	No	Yes	No	No	No	WD	MTMD
34	65	M	No	No	No	No	No	WD	MID
36	39	M	Yes	No	Yes	Yes	No	WD	MID
37	55	M	Yes	No	Yes	Yes	No	WD	MID
39	45	F	No	Yes	No	No	No	WD	MOD
42	53	M	No	No	No	Yes	Yes	WD	MTMD
43	70	M	Yes	No	No	No	No	WD	MOD
44	55	F	No	Yes	No	No	No	WD	MID
47	65	F	No	Yes	No	No	No	WD	SD
52	53	M	No	Yes	No	No	No	MD	MID

Table 2. Clinicopathological characteristics of 25 patients for whom RNAseq data was generated. Histopathological characterization terms are: *EI* early invasive, *MD* moderately differentiated, *WD* well differentiated tumour, *MID* mild, *MTMD* mild to moderate, *MOD* moderate, *SD* severe dysplasia.

in OSCC(median TPM expression: 0.186) despite reduced protein expression, implicating post-transcriptional or post-translational regulation, including microRNA silencing, protein degradation, or trafficking defects[36,37]. Similar discordance has been reported in breast cancer, where elevated DLG7 mRNA correlated with adverse prognosis independent of histological grade[38].

In OSCC, the loss of polarity proteins was widespread and uniform across tumor grades, mirroring experimental models in which disruption of the Scribble complex promotes unchecked proliferation. Interestingly, OSCC exhibited a modest restoration of polarity protein expression compared to OED. This partial re-expression aligns with a state of partial epithelial-to-mesenchymal transition (EMT), in which invasive cells maintain certain epithelial characteristics while simultaneously acquiring migratory capabilities. These data suggest that polarity loss reaches its nadir in dysplasia, priming cells for stromal invasion, whereas invasive carcinoma cells may rewire polarity programs to facilitate EMT-associated plasticity[35,39].

Collectively, our findings position polarity disruption as a fundamental and early driver of oral carcinogenesis. By delineating alterations in PAR3, SCRIBBLE, and DLG7, we identify robust biomarker candidates that enable objective dysplasia grading and risk stratification of OPMDs. Immunohistochemical assessment of these proteins provides a practical complement to routine histopathology, mitigating interpretative variability[15]. In OSCC, their differential expression corresponds with EMT and invasive behavior, highlighting prognostic value for tailoring treatment intensity. More broadly, polarity protein profiling offers a measurable biological endpoint for chemoprevention trials and a rationale for EMT-targeted therapies, thereby linking histopathological assessment with precision oral oncology[40]. While mechanistic and longitudinal validation remains essential, our findings elevate polarity rewiring from a descriptive histological feature to a molecular hallmark of malignant transformation in the oral epithelium.

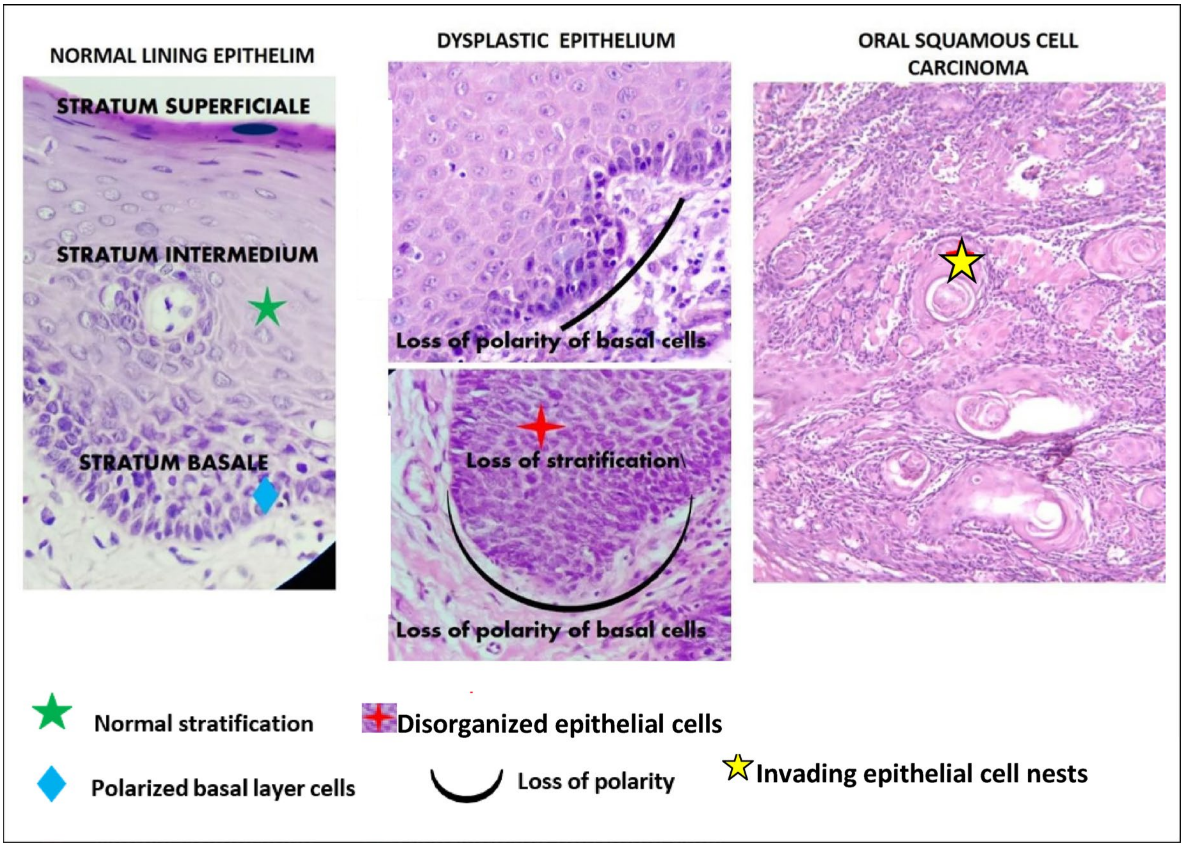

Fig. 1. H& E stained sections showing (from left to right) Normal oral mucosa (in the left) and its various stratified layers, Oral Epithelial dysplasia (in the middle) & its features and Well Differentiated Oral squamous cell carcinoma-WDSCC (in the right).

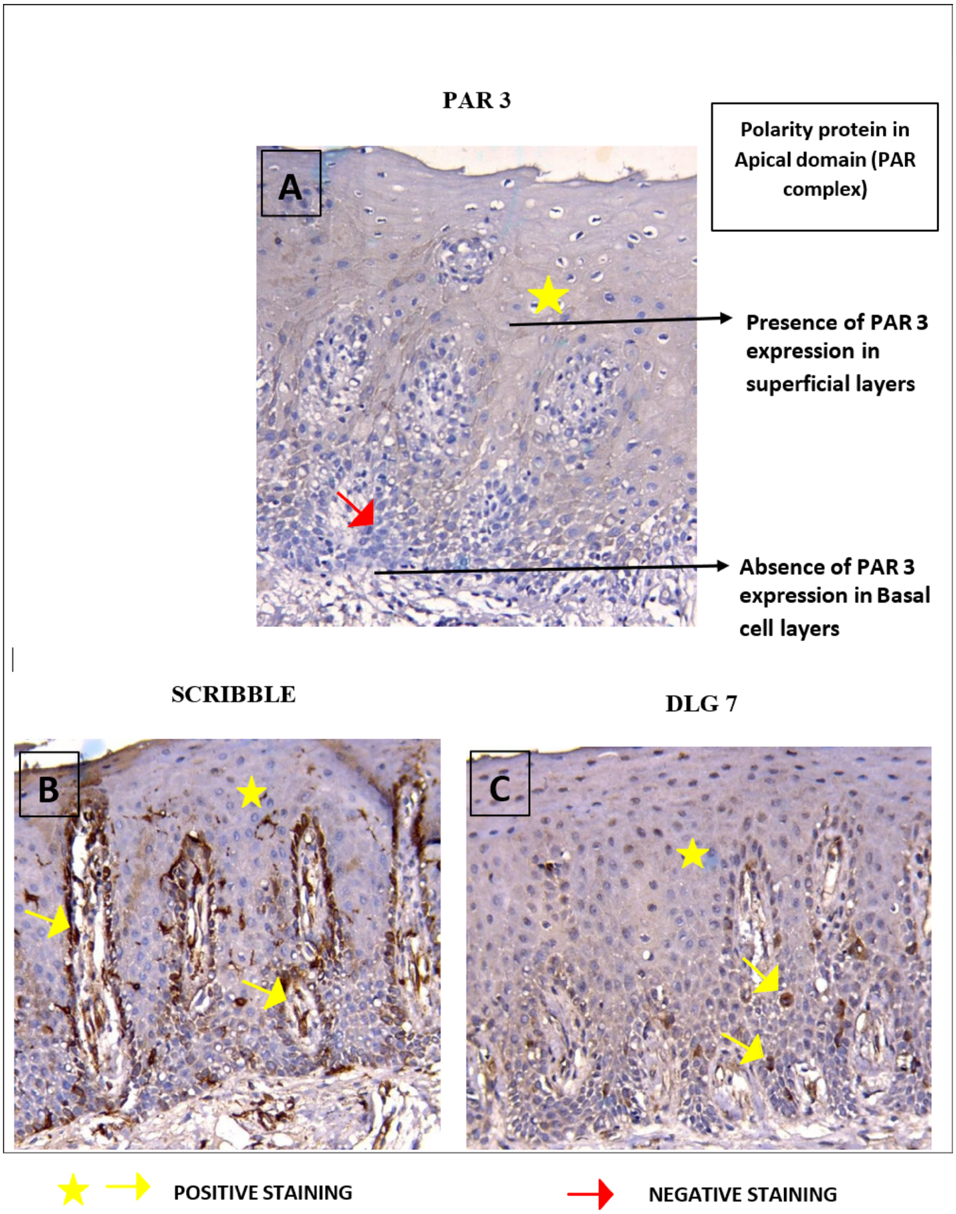

Fig. 2. Immuno-histochemical expression of PAR3 (**A**), Scribble (**B**), and DLG7 (**C**) expression in Normal Oral mucosa.

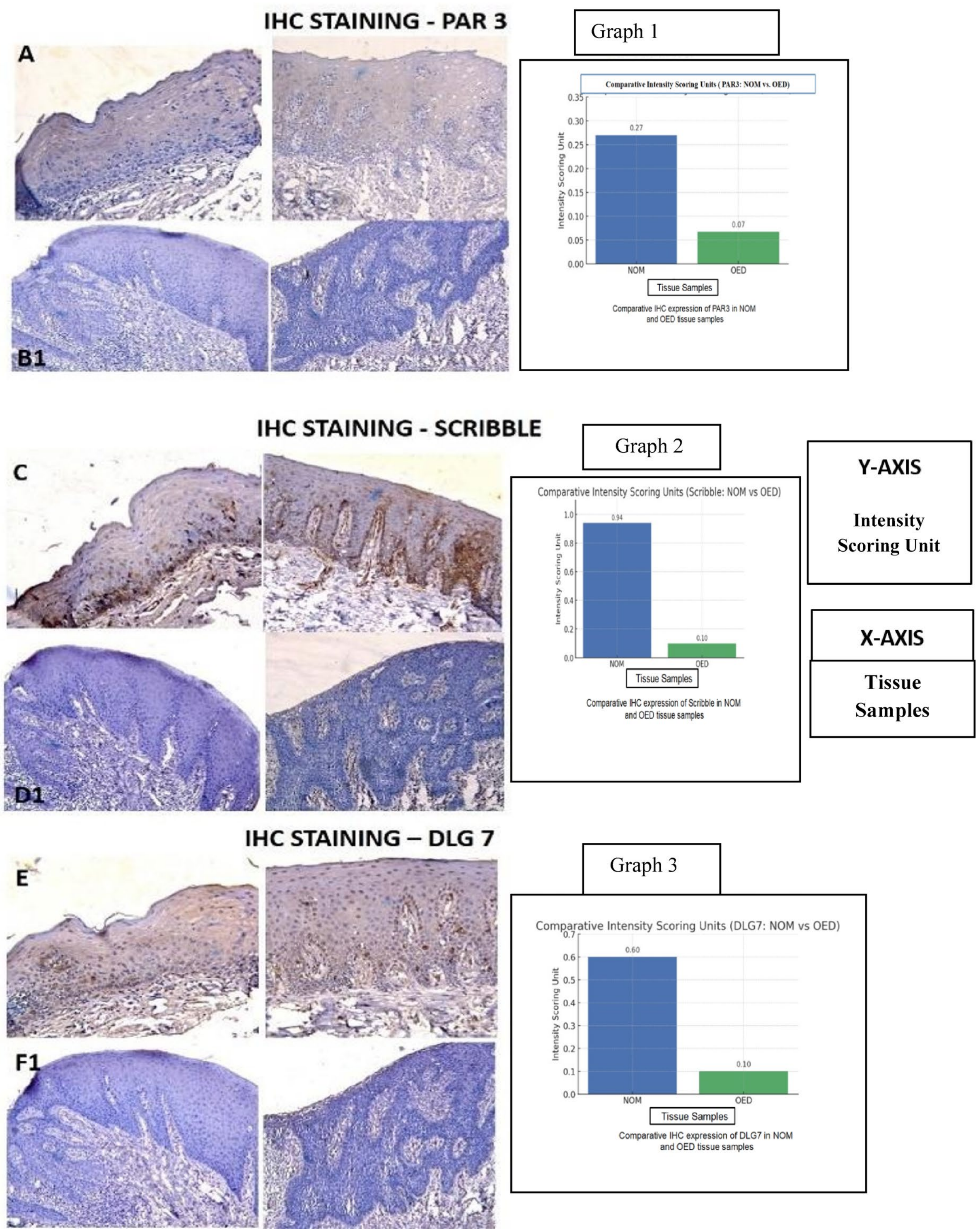

Fig. 3. Immunohistochemical staining of PAR 3, SCRIBBLE & DLG in normal oral mucosa (**A**,**C**,**E**) in comparison with OED (**B1**,**D1**,**F1**).

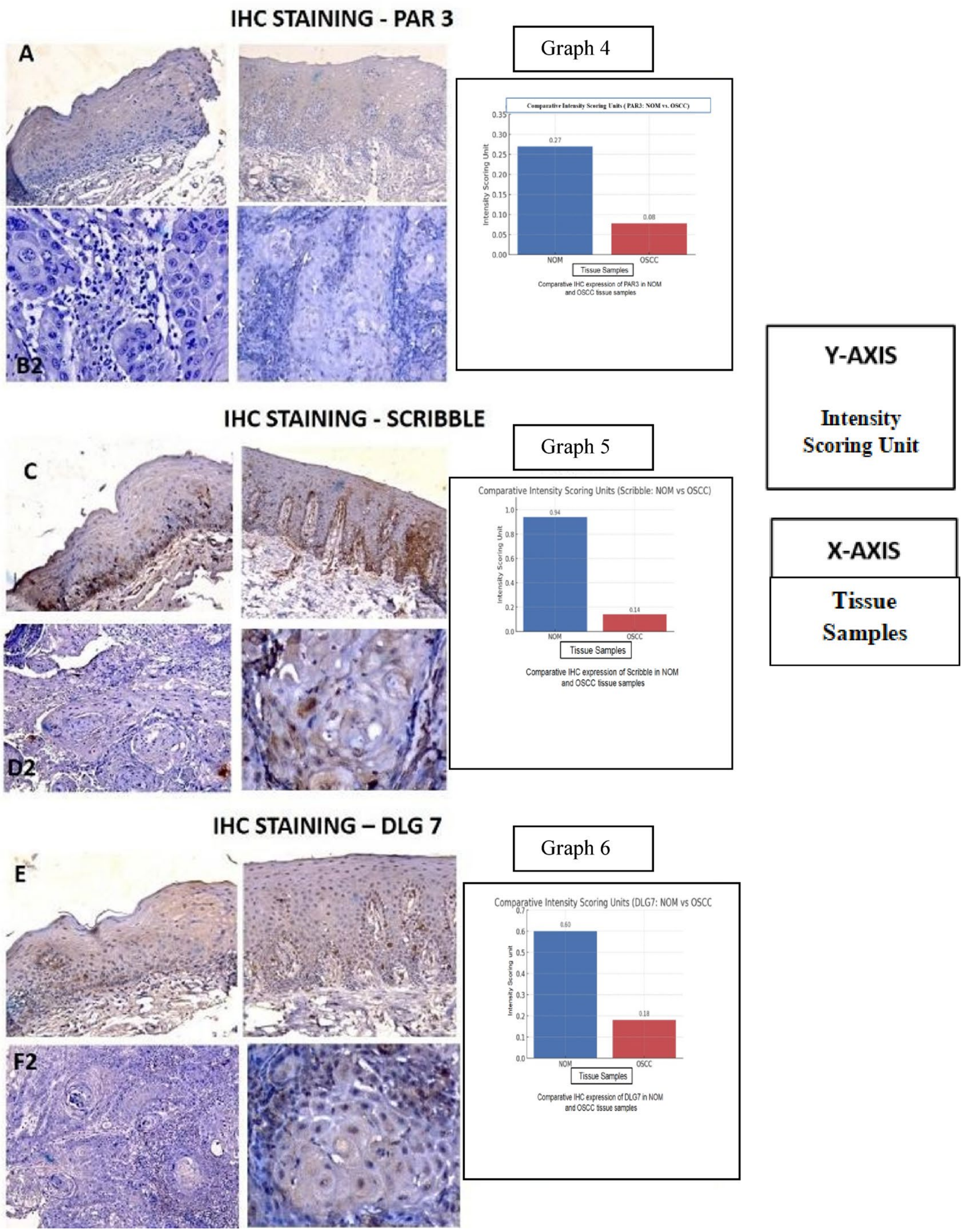

Fig. 4. Immunohistochemical staining of PAR 3, SCRIBBLE &DLG 7 in normal oral mucosa (**A**,**C**,**E**) comparison with OSCC (**B2**,**D2**,**F2**).

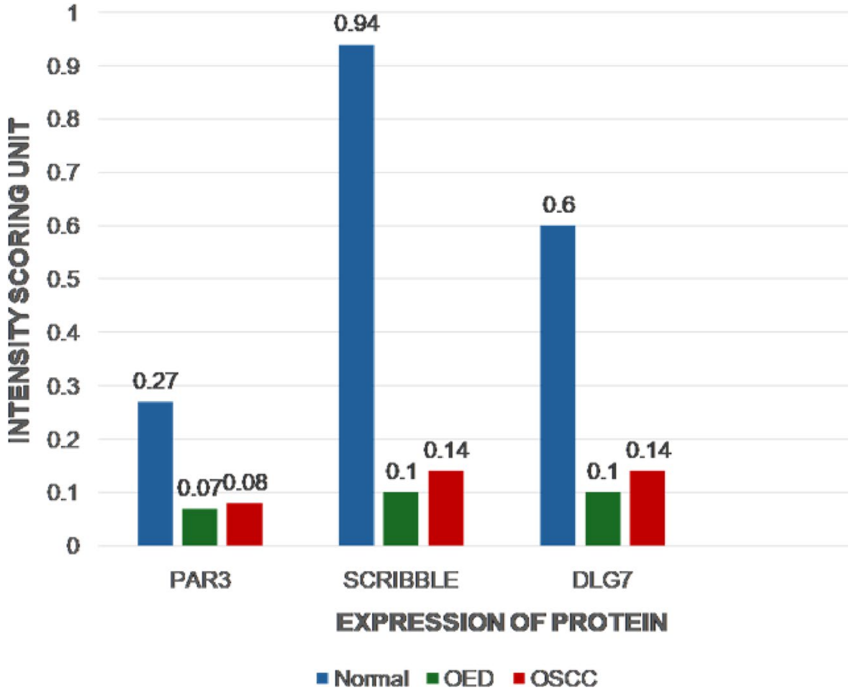

Fig. 5. Comparative IHC expression of PAR3, Scribble and DLG7 in NOM, OED and OSCC tissue samples.

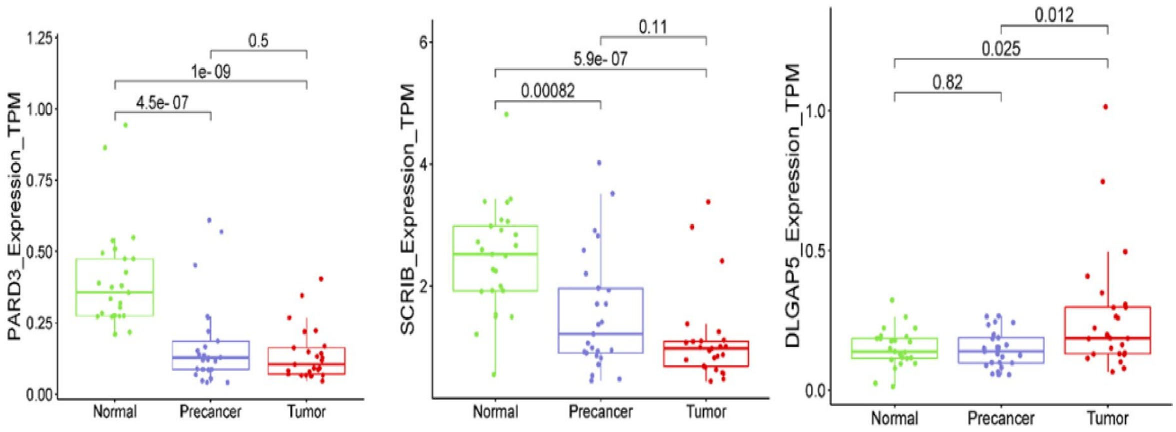

Fig. 6. Expression profile of PAR3, SCRIBBLE and DLG7 genes (from whole transcriptome, Normalized TPM expression values plotted) in oral precancer, tumor and normal samples.

Data availability

Transcriptome sequence data used in this study is publicly available and has been downloaded from the European Nucleotide Archive accession—PRJEB42969.

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Author contributions
SG conceived the study and performed the histopathological and immunohistochemical evaluations. URC handled patient recruitment and clinical assessments. SC and NKB generated and analyzed the transcriptomic data. SS and DR contributed to manuscript drafting. RNB provided overall guidance and coordinated the immunohistochemical investigations.

Declarations

Competing interests
The authors declare no competing interests.

Ethical approval
All the patients were included in the study after taking their Informed consents. All methods in this study were carried out in accordance with relevant Ethical guidelines and regulations. All experimental protocols were approved by the Institutional Ethics Committee (IEC) RADCH.

Additional information

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